

Research Statement

My research asks what a firm acquires when it buys technology from outside, and what that purchase changes inside the firm. A purchase is recorded as an addition to what the firm holds, and the papers below ask what follows from it, whether operations change, whether internal invention continues, and what the transaction does not transfer. The technology a firm holds and the environmental record it carries are not reported in financial statements. I measure both from text, primarily patent documents, regulatory disclosures, and corporate filings, and combine those measures with quasi-experimental variation in regulatory pressure, in adoption, and in what an acquirer knows before it bids.

Buying the Credential

A firm under environmental regulation is recorded twice, once by the regulator against the condition written into its permit and once by the audience that ranks firms on what they have in place. My job market paper, *“When Pressure to Be Green Becomes Demand for Green Credentials”*, joint with Kangryun Lee, asks whether regulatory pressure makes firms buy green technology that moves the second record. The design uses the 2005 designation of fine particulate nonattainment counties under the Clean Air Act. I construct the measure of exposure, the share of a firm’s pre-shock air emissions originating at plants in counties later designated. The obligation attaches to a plant through its location, so exposure is fixed by where a firm’s plants already stood.

Identifying a green acquisition requires knowing what technology the target owns. Prior work infers it from the acquirer’s stated motive or from the target’s industry classification. I read it from the target’s own granted green patents, classified through the World Intellectual Property Organization Green Inventory, so the label does not depend on what the acquirer says.

On a firm-year panel of 1,106 polluting firms reporting to the Toxics Release Inventory, I estimate that exposed firms become 4.08 percentage points more likely to acquire a target holding green technology, against a base rate of 2.86 percent, with no rise in overall acquisition activity. The increase is concentrated in climate and clean energy technology. Acquisitions of air-pollution abatement technology, which the permit requires, show no detectable response. We find that the increase is larger at firms that more analysts follow. After the deals, the firm’s environmental rating rises, emissions do not fall, and the acquirer’s own invention does not shift toward the acquired technology.

The targets and the patents are real, and the green label is accurate. Greenwashing is a gap between what a firm says and what it does. This paper finds a gap between what the firm holds and what the regulation was written to change. A design built to detect false statements does not see it.

Buying the Method

A second paper, *“AI CAPEX and Innovation: Evidence from GPU Adoption”*, joint with Raymond Kim, asks the same question of a firm buying AI computing capacity. A firm can obtain methods by buying capacity or by inventing them, and the paper asks what buying does to inventing.

We measure adoption at the firm level from graphics processing unit purchases and estimate a staggered difference-in-differences design. Patents are categorized into those that cover methods, known as process patents, and those that cover products. We find that adoption reduces internal process patenting. Total patenting, product patenting, research budgets, and inventor headcount are unchanged.

The reduction extends to methods unrelated to AI, where method patenting falls by a similar amount as in AI-related classes. The firm buys the methods it would have invented. It keeps the people and the budget that would have invented them.

The finding is the demand-side mirror of evidence that firms building AI capability internally produce more product innovation.

The Acquisition of Innovation

Two further projects examine what an acquisition of an innovative firm delivers. *“Choosing Twice: What the Contract Does Not Transfer in the Acquisition of Innovation”* begins from an asymmetry in the transaction. The purchase agreement conveys the target’s patents. The inventors who produced them are under no comparable obligation. Using corporate venture capital stakes as variation in what an acquirer knows before it bids, the project asks what prior information changes and what it does not.

“Try Before You Buy: Measuring Innovation Synergy Using Textual Analysis” is a measurement project. Innovation synergy in mergers is normally measured from firm-level patent counts and citations, which cannot see whether the two research organizations combined. I construct measures at the inventor level from patent text and track inventors by identity after the organizations merge, so that the combination is observed where it would occur. The project has been presented at SFA 2025 and MFA 2026. A related project, *“Common Ownership of Corporate Venture Capital and the Economic Value of Innovation”*, received the Mays Innovation Research Center Ph.D. Paper Award.

Outside these lines, *“Who Mimics? Leaders, Followers, and the Decline of Capital Structure Peer Effects”* extends Leary and Roberts (2014) to a 1975 to 2023 sample and documents that peer effects in capital structure are concentrated among industry followers, with leaders showing no measurable response, and that the post-2000 decline in the aggregate effect reflects the entry of firms that do not mimic.

Data and Future Direction

I built the data infrastructure behind these papers. It links patent full text, classifications, and inventor records to firms across VentureXpert, SDC, Crunchbase, and Compustat, and joins facility-level records from the Environmental Protection Agency Toxics Release Inventory to parent firms. The text-based measures include the classification of green technology through the World Intellectual Property Organization Green Inventory, cosine similarity and term frequency inverse document frequency over patent text, and keyword overlap at the inventor level. I work in Stata, R, Python, and L^AT_EX.

Future work runs in three directions. The first is artificial intelligence and financial technology, the second sustainable finance, and the third machine learning and textual analysis as measurement methods.

These datasets and methods extend readily to questions colleagues bring to them, and I would welcome coauthors and doctoral students working in these areas.